


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
AIOT
21-04-2024
JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

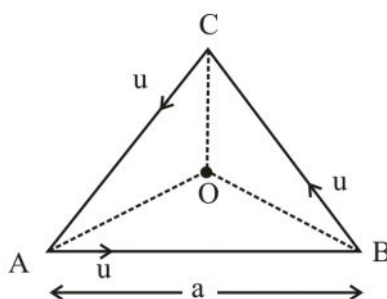
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- Three particles A, B and C situated at vertices of an equilateral triangle, all moving with same constant speed such that A always move towards B, B always towards C and C always towards A. Initial separation between each of the particle is a . O is the centroid of the triangle. Distance covered by particle A when it completes one revolution around O is

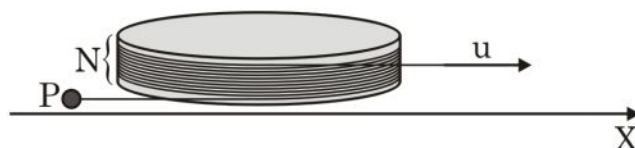


- $2a \left(1 - e^{-2\sqrt{3}\pi} \right)$
 - $\frac{2a}{3} \left(1 - e^{-2\sqrt{3}\pi} \right)$
 - $a \left(1 + e^{-2\sqrt{3}\pi} \right)$
 - $\frac{2a}{3} \left(1 - e^{-\sqrt{3}\pi} \right)$
- Use a diameter 1 mm superconductive wire to construct a circle of radius 5 cm. When the circular loop is superconductive, the current is 100 A. After one year, it is observed that the circular loop's change in current is less than 10^{-6} A. The resistivity of wire (in $\Omega\text{-m}$) is of the order of :-
 - 10^{-20}
 - 10^{-25}
 - 10^{-30}
 - 10^{-35}

3. In cloudy weather, measurements are taken with a light meter pointed towards the sky. We find that, due to the scattering of light by the clouds, the incident power per unit area is approximately I_0 , regardless of the orientation of the light meter. On top of an opaque thin walled spherical shell of radius R , there is a small hole of radius r (much larger than the wavelength of visible light). The inner side of the wall of the spherical shell is coated with soot. The intensity of illumination on the inner surface of the shell, when it is placed in the open air is :-

- (A) $I_0 \frac{r^2}{R^2}$
 (B) $I_0 \frac{r^2}{4R^2}$
 (C) $I_0 \frac{r^2}{R^2 - r^2}$
 (D) $I_0 \frac{r^2}{R^2 + r^2}$

4. A massless thread makes N turns around statically fixed cylinder, as shown in the figure. Initially, the free (unwound) ends of the thread are parallel to the axis X . Then, a heavy point-like object P is attached to one end of the thread while the other end is pulled with a constant velocity u along X . Find the velocity attained by the heavy object when it detaches from the cylinder. Suppose that the turns of the thread are wound tightly to one another and are placed practically in the same plane, perpendicular to the cylinder axis. Neglect any friction in the system. Do not consider the force of gravity.



- (A) u
 (B) $2\pi Nu$
 (C) $\sqrt{1 + (2\pi N)^2} u$
 (D) πNu

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

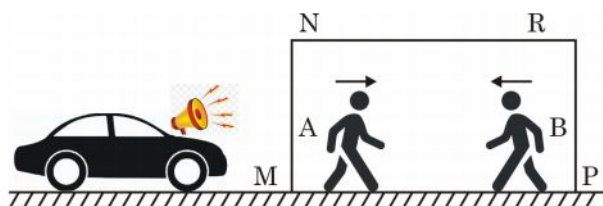
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

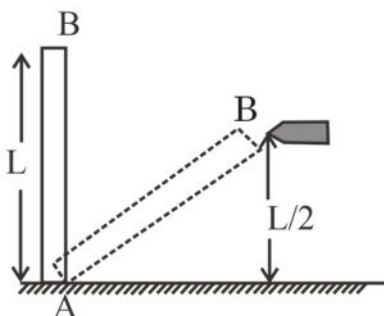
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A car is moving with a velocity $\frac{3v}{10}$ towards a medium MNRP as shown, where 'v' is the speed of sound in air. Two men A and B are moving towards each other in the medium with the velocities $\frac{v}{6}$ and $\frac{v}{9}$ respectively. The speed of sound in medium is $7v$ and medium is kept at rest. The car is emitting a sound of frequency f . Then choose the correct option(s).

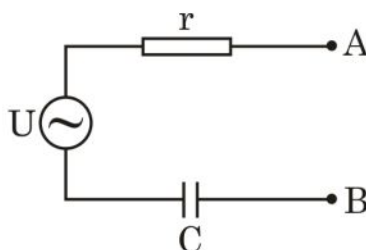


- (A) The wavelength received by the man A is $\frac{49v}{10f}$
- (B) The wavelength received by the man A is different from the wavelength received by man B
- (C) The ratio of frequency observed by the man A to that observed by the man B is $\frac{123}{128}$
- (D) The frequency observed by the man B is $\left(\frac{640}{441}\right) f$

6. A rod of mass 'M' and length 'L' was at rest in a vertical position, its end A being hinged about which it can freely rotate. A slight disturbance caused the rod to start rotating clockwise. A bullet of mass 'm' and speed ' v_0 ' flying horizontally at a height ' $L/2$ ' above ground hits the end B of the rod and gets embedded. (Take $M = 3m$). Then choose the correct statement(s).



- (A) Minimum velocity of the bullet such that the rod again becomes vertical is $v_0 = \sqrt{gL} (\sqrt{6} + \sqrt{20})$
- (B) Tangential acceleration of the midpoint of the rod when it makes an angle 60° with horizontal after collision is $\frac{5g}{16}$
- (C) Minimum velocity of the bullet such that the rod again becomes vertical is $v_0 = \sqrt{3gL} (\sqrt{6} + \sqrt{20})$
- (D) Tangential acceleration of the midpoint of the rod when it makes an angle 60° with vertical after collision is $\frac{5g}{12}$
7. The equivalent circuit of a real source of alternating voltage of frequency $\omega = 1.00 \times 10^3 \text{ s}^{-1}$ consists of an ideal voltage source with the amplitude $U = 15.0 \text{ V}$, a resistor with the resistance $r = 10\Omega$ and a capacitor with the capacitance $C = 100 \mu\text{F}$. Different circuits of resistors, capacitors and coils can be connected to the source as a load. A load circuit, which assures maximum heat generated in the load itself is connected across the source terminals AB.



- (A) The load is purely resistive in nature.
- (B) The power factor of the load is $\frac{1}{\sqrt{2}}$.
- (C) The resistance of the load is equal to 10Ω .
- (D) The maximum heat generated in the load is $\frac{45}{32} \text{ W}$.

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A solid cylinder of mass m and radius r rests on the inside of a thin-walled cylinder of mass M and radius R . The solid cylinder is made to oscillate around the equilibrium position.

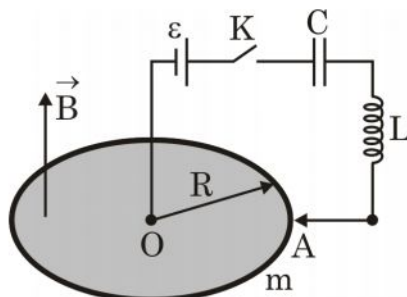
(Take $M = 1\text{ kg}$, $m = 1\text{ kg}$, $R = 1\text{ m}$ and $r = 0.25\text{ m}$)

1. The angular frequency (in rad/s) of oscillation if the large cylinder is fixed and small cylinder rolls without slipping in the bottom of the cylinder is :-
2. The angular frequency (in rad/s) of oscillation if the large cylinder is free to rotate about an axis through its fixed centre and small cylinder rolls without slipping in the bottom of the cylinder.

Paragraph for Questions 3 and 4

The figure shows a diagram of an electrical circuit consisting of a source with electromotive force ε , a capacitor with capacitance C , a coil of inductance L , a key K and a metal disk of radius R . The disk can rotate around its axis without friction. The disk is located in a uniform magnetic field with induction B , directed along its axis as shown in the figure. Almost the entire mass of the disk is concentrated in its thin rim. The source is connected to the disk at its centre O , and the inductor is connected to the disk using a fixed sliding contact A . Initially, key K is open, the disk is motionless, there is no current in the coil, and capacitor is not charged. Key K is closed.

(Take : $B = 1\text{ T}$, $C = 1\text{ F}$, $\varepsilon = 1\text{ V}$, $m = 1\text{ kg}$, $R = 1\text{ m}$, $L = 1\text{ H}$, $\pi = 3.14$ and $\sqrt{5} = 2.24$)



3. The maximum angular velocity (in rad/sec) of the disc after key is closed is :-
4. The time (in sec) after closing the key after which the angular velocity of the disk first reaches the maximum value is :-

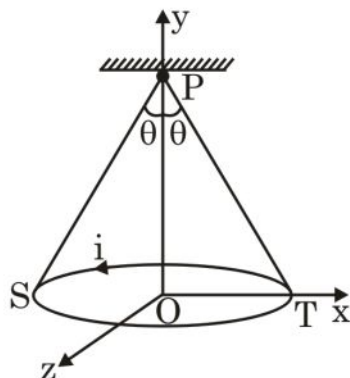
SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

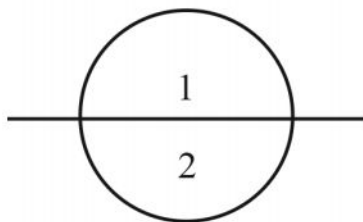
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

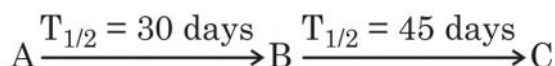
5. A ring of uniform mass density of mass m ($m = 100 \text{ gm}$) and radius R ($R = 50 \text{ cm}$) is suspended by two insulated strings PS and PT from the point P on y-axis in vertical plane. The ring is kept parallel to horizontal surface on x-z plane. Ring carries a current ($i = \frac{7}{22} \text{ amp}$) as shown in the figure. There exist a magnetic field ($B = 1 \text{ Tesla}$) along negative x-axis. The tension in the string PS is T_1 and in the string PT is T_2 . Find the ratio $\frac{T_1}{T_2}$. (It is given that $\theta = 60^\circ$)



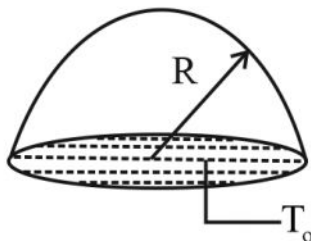
6. If we consider earth as a uniform solid sphere of mass M and Radius R then force of interaction between two hemi-spheres will be $\frac{3GM^2}{\alpha R^2}$. Find α



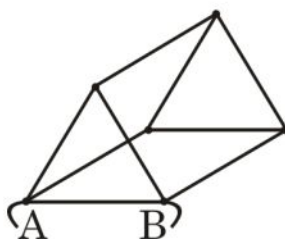
7. A given sample contains two types of atoms A and B in the ratio 3 : 1. Atoms of type A undergo α -decay with a half life of 30 days to form 'B' while atoms of type B undergo α -decay with a half life of 45 days to form 'C', which is stable. Calculate the time (in days) after which the activities of A and that of B are in the ratio 9 : 22. If time is T write $\frac{T}{10}$.



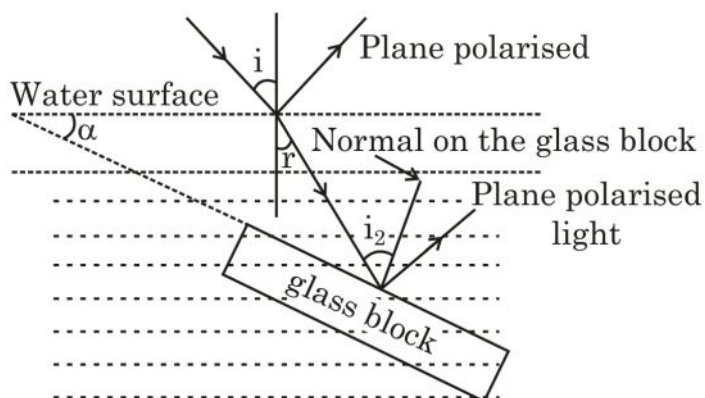
8. Circular base of hemi-spherical shell base closed body is maintained at $T_0 = 30\text{K}$, outer atmosphere is at 0 K . If equilibrium temperature of curved surface is $10 \times (x)^{\frac{3}{4}}\text{K}$, write x :-



9. The resistance between two joints is R . The equivalent resistance between A and B is $\frac{\alpha R}{15}$. The value of α is :-



10. Unpolarised light is incident on a water surface at the Brewster angle. The refracted ray meets the block of glass $\left[\mu_g = \frac{3}{2}\right]$ immersed in the water $\left(\mu_w = \frac{4}{3}\right)$. The flat upper surface of the block makes an angle α with water surface. The value of α for which the light reflected from the block is completely polarized. $(\tan^{-1} \frac{9}{8} = 48.4)$ Fill the value of 10α .



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

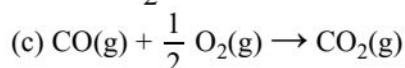
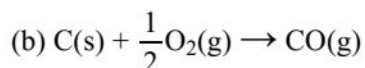
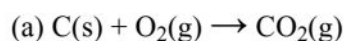
- This section contains **FOUR (04)** questions.
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- For each question, choose the option corresponding to the correct answer.
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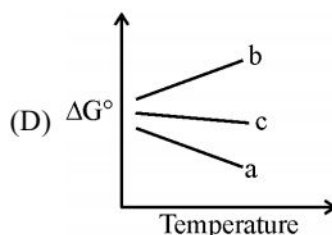
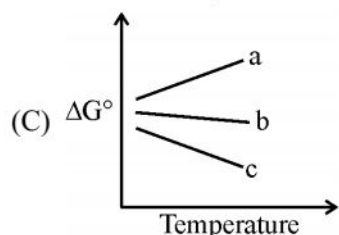
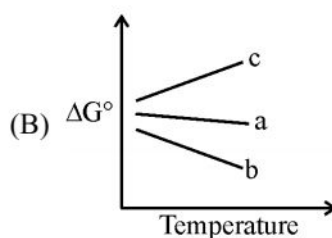
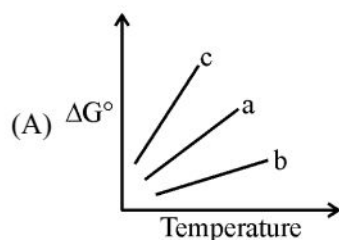
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. The standard Gibb's free energy of the following reaction, ΔG° have been determined at various temperatures :



Which of the following plots would represent most likely the temperature dependence of ΔG° ?



2. Potential energy of electron present in 2nd orbit of Li^{2+} is : (a_0 = Radius of 1st Bohr's orbit) :

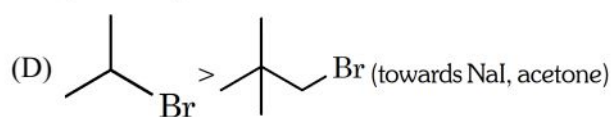
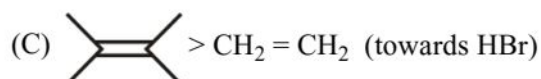
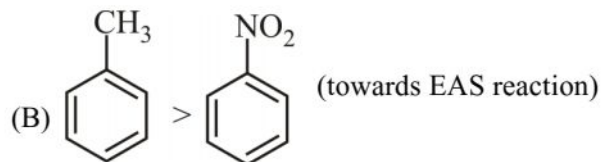
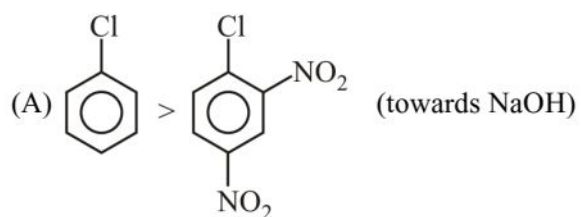
(A) $-\frac{e^2}{4\pi\epsilon_0 a_0}$

(B) $-\frac{3e^2}{4\pi\epsilon_0 a_0}$

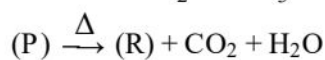
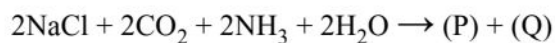
(C) $-\frac{3e^2}{16\pi\epsilon_0 a_0}$

(D) $-\frac{9e^2}{16\pi\epsilon_0 a_0}$

3. Which of the following is not correct order of reactivity.



4. Consider the following reaction/s & choose **INCORRECT** statement.



(A) In solid salt (P) anion form an infinite chain

(B) From compound (Q) NH_3 can be obtained by adding $\text{Ca}(\text{OH})_2$

(C) Compound (P) has symmetrical H-bond

(D) MnO_2 salt give green coloured bead when heated with compound (R) in presence of air.

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

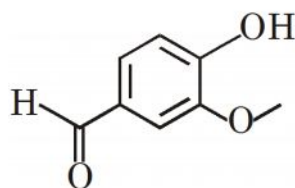
5. At 298 K, standard reduction potential is 1.33 V for $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}$.

Select the correct statement(s). (If concentration of other species are 1M)

Given : $\frac{2.303R \times 298}{F} = 0.06 \text{ V}$, $E^\circ_{\text{Br}_2/\text{Br}^-} = 1.07 \text{ V}$ and $E^\circ_{\text{I}_2/\text{I}^-} = 0.54 \text{ V}$.

- (A) $\text{Cr}_2\text{O}_7^{2-}$ may oxidise Br^- and I^- at $\text{pH} = 3$
 (B) $\text{Cr}_2\text{O}_7^{2-}$ may oxidise Br^- and I^- at $\text{pH} = 0$
 (C) $\text{Cr}_2\text{O}_7^{2-}$ may oxidise I^- but can't oxidise Br^- at $\text{pH} = 3$
 (D) $\text{Cr}_2\text{O}_7^{2-}$ may oxidise I^- but can't oxidise Br^- at $\text{pH} = 0$

6. Choose the **CORRECT** statement(s) regarding Vanilin



Vanilin [P]

- (A) [P] can be obtained as a major product when 3,4-dihydroxy benzaldehyde is treated with one equivalent of NaOH and one equivalent of MeBr.
- (B) [P] can be obtained as a one of the product when 2-methoxy phenol is treated with chloroform and aq KOH.
- (C) Positional isomer of [P] can be obtained when 3,4-dihydroxy benzaldehyde is treated with one equivalent of NaOH followed by treatment with one equivalent of MeBr.
- (D) [P] can be obtained when 3,4-dihydroxy benzaldehyde is treated with two equivalent of NaOH and one equivalent of MeBr followed by protonation.
7. Choose correct statement/s about extraction of various metal in metallurgical process.
- (A) In Mond process the decomposition of nickel carbonyl is accompanied by a large entropy increase (four moles of gas formed from one mole of complex) and hence it takes place spontaneously at higher temperature when $T\Delta S$ term becomes significant
- (B) Crude gold obtained by Mac Arthur Forest cyanide process contain Zn and sometime Pb impurity
- (C) In Van Arkel-De Boer process the impure element is heated with iodine, producing a volatile iodide.
- (D) In Kroll process
- $$\text{TiCl}_4 + 2\text{Mg} \xrightarrow{\text{under atmosphere of Ar}} \text{Ti} + 2\text{MgCl}_2$$
- Here MgCl_2 is leached with H_2O

SECTION-II (i) : (Maximum Marks: 12)

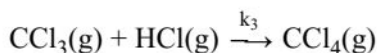
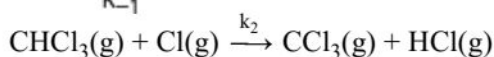
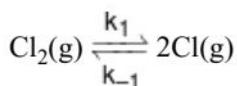
- This section contains **TWO (02)** paragraphs.
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- The answer to each question is a **NUMERICAL VALUE**.
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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

For the reaction, $\text{CHCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \xrightarrow{h\nu} \text{CCl}_4(\text{g}) + \text{HCl}(\text{g})$ mechanism is given below



Given :

$$k_1 = 4.8 \times 10^3 \text{ sec}^{-1}, k_{-1} = 1.2 \times 10^3 \text{ M}^{-1} \text{ sec}^{-1},$$

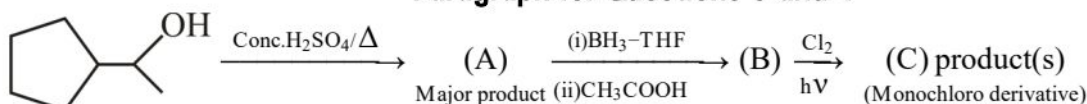
$$k_2 = 1.3 \times 10^{-2} \text{ M}^{-1} \text{ sec}^{-1}, k_3 = 2.1 \times 10^2 \text{ M}^{-1} \text{ sec}^{-1}$$

$$E_{a1} = 10 \text{ kJ/mol}, E_{a-1} = 15 \text{ kJ/mol},$$

$$E_{a2} = 45 \text{ kJ/mol}, E_{a3} = 12 \text{ kJ/mol}$$

1. Overall rate constant of reaction, $\text{CHCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \xrightarrow{h\nu} \text{CCl}_4(\text{g}) + \text{HCl}(\text{g})$, is $\text{_____} \times 10^{-2} \text{ M}^{-1/2} \text{ sec}^{-1}$.
2. Overall activation energy of reaction, $\text{CHCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \xrightarrow{h\nu} \text{CCl}_4(\text{g}) + \text{HCl}(\text{g})$, is _____ kJ/mole .

Paragraph for Questions 3 and 4



3. Total number of fractions obtained on fractional distillation of all monochloro derivatives product(s) (C)?
4. Calculate mass of Cl_2 required to react with 9.8 gm of compound (B) to form monochloro product. Assuming % yield = 100%

SECTION-II (ii) : (Maximum Marks: 24)

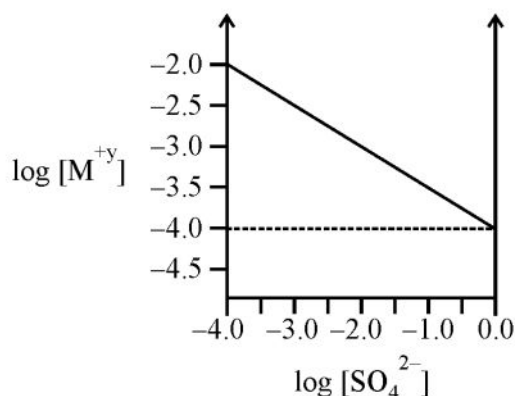
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

5. A metal M form sparingly soluble metal sulphate. The logarithm of concentration of dissolved metal ion in a solution saturated in the metal sulphate is plotted below as a function of logarithm of concentration of sulphate ion in solution. The molecular formula of metal sulphate is $M_x(SO_4)_y$. Then find $z \times 10^{-6}$.

$$z = \frac{x + y}{K_{sp} \text{ of } M_x(SO_4)_y}, \text{ x and y are minimum integer co-efficient.}$$



6. Total number of Pseudohalides ion which are planar and polar.
 $CN^\ominus$, $SCN^\ominus$, $SeCN^\ominus$, $OCN^\ominus$, CN_2^{2-} , $ONC^\ominus$, $N_3^\ominus$, $I_3^\ominus$, $ICl_4^\ominus$
7. Total number of reaction/s in which paramagnetic gaseous product form on heating.
- (1) $KMnO_4 \xrightarrow{\Delta}$
 - (2) $MnO_2 + \text{conc. HCl} \xrightarrow{\Delta}$
 - (3) $MnO_2 + \text{conc. H}_2\text{SO}_4 \xrightarrow{\Delta}$
 - (4) $KClO_3 \xrightarrow{\Delta}$
 - (5) $Pb(NO_3)_2 \xrightarrow{\Delta}$
 - (6) $CaCO_3 \xrightarrow{\Delta}$
 - (7) $NaNO_3 \xrightarrow{\Delta}$
 - (8) $NH_4NO_3 \xrightarrow{\Delta}$
 - (9) $Na_2CrO_4 + \text{conc. H}_2\text{SO}_4 \rightarrow$

8. Total number of stereoisomer possible for complex compound $\left[(\text{en})_2\text{Co} \begin{array}{c} \diagup \text{NO}_2 \diagdown \\ \diagdown \text{NH}_2 \diagup \end{array} \text{Co}(\text{en})_2 \right]^{4+}$ is.

9. $\text{NaCl}_{(\text{s})} + \text{K}_2\text{Cr}_2\text{O}_{7(\text{s})} + \text{conc. H}_2\text{SO}_4 \xrightarrow{\text{Heat}}$ Reddish brown fumes
 $\downarrow$
 $\text{NaOH}_{(\text{aq.})}$
 Yellow solution (A)

Total number of reagent/s which give redox reaction with solution (A) in acidic medium among the following.

$\text{Pb}(\text{CH}_3\text{COO})_2$, $\text{Ba}(\text{NO}_3)_2$, NO_2^- , Sn^{2+} , SO_2 , Fe^{2+} , H_2S , I^- , $\text{C}_2\text{H}_5\text{OH}$

10. How many amino acids are positively charged at pH = 7.
- | | | |
|---------------------|-----------------|---------------------|
| (i) Asparagine | (ii) Valine | (iii) Aspartic acid |
| (iv) Glycine | (v) Lysine | (vi) Glutamine |
| (vii) Glutamic acid | (viii) Arginine | (ix) Methionine |

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- Let $\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \left(\sum_{r=1}^n \ell_n (n^2 + r^2) \right) - 2n \ell_n n \right\} = \ell_n 2 + \frac{\pi}{2} - 2$
If $\lim_{n \rightarrow \infty} \frac{1}{n^{2m}} \left[(n^2 + 1^2)^m (n^2 + 2^2)^m (n^2 + 3^2)^m \dots (n^2 + n^2)^m \right]^{\frac{1}{n}} = \left(\frac{a\sqrt{e\pi}}{e^2} \right)^m$, then the value of a is
(where $m, n \in \mathbb{N}$)
(A) 1
(B) $\frac{3}{2}$
(C) 2
(D) $\frac{1}{2}$
- The remainder when $\sum_{k=0}^{1012} \left[\frac{{}^{1012}C_k}{{}^{2024}C_k} \sum_{r=k}^{2024} {}^rC_k \cdot {}^{2024}C_r \right]$ is divided by 25 is (where ${}^nC_r = \frac{n!}{r!(n-r)!}$)
(A) 15
(B) 17
(C) 21
(D) 11

3. A moving particle starts at the point (4, 4) and moves until it hits one of the coordinate axes for the first time. When the particle is at the point (a, b) it moves at random to one of the points (a - 1, b), (a, b - 1) or (a - 1, b - 1) each with probability $\frac{1}{3}$, independent of its previous moves. The probability that it will hit the coordinate axes at (0, 0) is $\frac{x}{3^y}$. Where x and y are positive integers such that x is not divisible by 3. Then value of x + y is
- (A) 252
(B) 253
(C) 152
(D) 353
4. Let H be the hyperbola having eccentricity 'e' and C be the smallest circle touching H and its pair of asymptotes. Ratio of radius of C with the length of semi-conjugate axis of H is.
- (A) $e + \sqrt{e^2 - 1}$
(B) $e - \sqrt{e^2 - 1}$
(C) $e - \sqrt{e^2 + 1}$
(D) $\frac{1}{e}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
5. Let A and B be two 3×3 matrices where both A and $I + AB$ are non-singular. If $3A^2B^2 + AB = 2(AB)^2$ then, which of the following is greater than $|B|$ (where I is identity matrix of order 3 & $|X|$ denotes determinant of X)
- (A) 1
(B) 2
(C) 3
(D) 4

6. Consider $f : \left[0, \frac{\pi}{2}\right] - \left\{\frac{\pi}{4}\right\} \rightarrow \mathbb{R}$. Let $f(x) = \lim_{n \rightarrow \infty} \prod_{k=0}^n \frac{((\cot x)^{2^k} + 1)^2}{((\cot x)^{2^{k+1}} + 1)}$, $\forall x \in \left(0, \frac{\pi}{2}\right] - \left\{\frac{\pi}{4}\right\}$ and $f\left(\frac{\pi}{4}\right) = 2$ and $f(0) = 0$, Then which of the following is true?
- (A) $f(x)$ is increasing in $\left(0, \frac{\pi}{4}\right)$ & $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
- (B) $f(x)$ is decreasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
- (C) $f(x)$ has one local maxima
- (D) $f(x)$ has three local minima
7. If $z = e^{i\pi/13}$ then $\frac{1}{1-z}$ is equal to (Where $i = \sqrt{-1}$)
- (A) $1 - z^2 + z^4 - z^6 + z^8 - z^{10} + z^{12}$
- (B) $1 + z^2 + z^4 + z^6 + z^8 + z^{10} + z^{12}$
- (C) $z - z^3 + z^5 - z^7 + z^9 - z^{11}$
- (D) $z + z^3 + z^5 + z^7 + z^9 + z^{11}$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider a circle $C : x^2 + y^2 = 5$,

Plane $\pi_1 : x - y + 2z + 5 = 0$ and

Plane $\pi_2 : 2x + y + z - 5 = 0$ and

Plane $\pi_3 : x + 2y - z = 0$.

Let reflection of curve 'C' about plane mirror π_1 be curve C_1 . [.] represents G.I.F.

1. If λ is the shortest distance of plane π_2 from any point on C_1 then $[\lambda]$ equal to:
2. Planes π_1, π_2, π_3 , form a prism. If 'A' is area of cross-section of this prism then [A] equal to

Paragraph for Questions 3 and 4

Consider $\int_0^{\frac{\pi}{2}} (\sin x)^m (\cos x)^n dx, m, n \in \mathbb{N}$.

We can simplify this integration by substituting $\sin x = t$ OR $\cos x = t$.

Based on this solve below questions

3. $I_1 = \int_0^1 x^6 (1-x)^4 dx, I_2 = \int_0^1 x^5 (1-x)^3 dx$

then $\frac{I_1}{I_2} = \frac{p}{q}$, where p, q are co-prime positive integers then $p + q$ equal to

4. Let $I = \int_0^{\frac{\pi}{4}} \frac{\left(\cos\left(x + \frac{\pi}{4}\right)\right)^{10} \cdot (\sin x)^7}{(\cos x)^{19}} dx = \frac{m!n!}{2^p \cdot 18!}$, m, n, p are natural numbers then smallest value of

$m + n + p = ?$

SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
5. Let us consider a complex number $z = x^{3/2} + iy^2$; $x \geq 0$; $y \in \mathbb{R}$ such that $2x^5y - x^4y + 2x^3y - x^2y + 4xy - 2y = 0$ and $|z|^2 = (\operatorname{Re}(z))^{2/3}$. Then number of all distinct value(s) of $|z|$ is
 6. Consider a circle $S : x^2 + y^2 + x - y = 2021$. If (x, y) is called integral points if x and y are both integer. Then number of integral points on $S = 0$ is
 7. Find number of permutations x_1, x_2, x_3, x_4, x_5 of numbers 1, 2, 3, 4, 5 taken all at a time such that the sum of five products $x_1x_2x_3 + x_2x_3x_4 + x_3x_4x_5 + x_4x_5x_1 + x_5x_1x_2$ is divisible by 3.
 8. The third derivative of $\frac{x^4}{(x-1)(x-2)}$ is $-\frac{k}{(x-2)^4} + \frac{6}{(x-1)^4}$. The number of $\theta \in [0, 2\pi]$ satisfying the equation $\sin^2\theta + 2\cos^2\theta = k - 96 - 3\sin\theta \cos\theta$ is ____.
 9. If $A = \begin{bmatrix} 6 & 3 \\ 2 & 1 \end{bmatrix}$ and $10^3 < \operatorname{tr}(A^n) < 10^4$ where $n \in \mathbb{N}$. The value of n is (where $\operatorname{tr}(X)$ denotes trace of matrix X)
 10. Let us consider the curves $9x^2 + 4y^2 = 900$, $x^2 + y^2 - 4x - 6y - 12 = 0$ and $y^2 - 6y - 6x + 51 = 0$. The maximum number of disjoint regions into which these curve divide xy plane (excluding themselves).

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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